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3D environment reconstruction for persistent object tracking

Abstract

In various examples, a 3D representation of an environment may be generated from sensor data, with objects being detected in the environment using the sensor data and stored as items that can be tracked and located within the 3D representation. The 3D representation of the environment and item information may be used to determine (e.g., identify or predict) a location or position of an item within the 3D representation and/or recommend a storage location for the item within the 3D representation. Using a determined location or position, one or more routes to the location through the 3D representation may be determined. Data corresponding to a determined route may be provided to a user and/or device. User preferences, permissions, roles, feedback, historical item data, and/or other data associated with a user may be used to further enhance various aspects of the disclosure.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11010961	12/2020	Stachniak	N/A	G06N 3/08
2006/0112034	12/2005	Okamoto	706/16	G05D 1/0274
2011/0235898	12/2010	Watanabe	382/154	G06T 7/33
2015/0290795	12/2014	Oleynik	700/257	G05B 19/42
2017/0072558	12/2016	Reynolds	N/A	G05D 1/024
2019/0086549	12/2018	Ushani	N/A	G05D 1/0088
2021/0279957	12/2020	Eder	N/A	G06V 20/64
2021/0397645	12/2020	Wang	N/A	G06F 18/22
2022/0245886	12/2021	Liebllich	N/A	G06T 15/08
2022/0327608	12/2021	Assouline	N/A	G06T 19/006
2022/0405874	12/2021	Barbour	N/A	G06V 10/762
2023/0150429	12/2022	Slutsky	345/419	H04N 13/271

OTHER PUBLICATIONS

What is YOLO Algorithm? | Deep Learning Tutorial 31 (Tensorflow, Keras & Python). codebasics YouTube Channel. <https://www.youtube.com/watch?v=ag3DLKsl2vk>. Posted Dec. 25, 2020. cited by examiner

Liu, et al.; "Furnishing Your Room by What You See: An End-to-End Furniture Set Retrieval Framework with Rich Annotated Benchmark Dataset," Retrieved from: arXiv: 1911.09299; Jan. 21, 2020, 12 pgs. cited by applicant

Bansal, et al.; "Marr Revisited: 2D-3D Alignment via Surface Normal Prediction," Retrieved from: arXiv:1604.01347; Apr. 5, 2016, 11 pgs. cited by applicant

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Background/Summary

BACKGROUND

(1) Conventional systems for tracking and assisting users in locating items in an environment—such as a home, a warehouse, an office, or a store—typically rely on a tracking device affixed to each item of interest. The tracking devices may be implemented using Bluetooth, global system for mobile Communications (GSM), global positioning system (GPS), or radio frequency identification (RFID). While each of these technologies are suitable for tracking items in various scenarios, they are generally limited in their ability to precisely track and/or indicate the location of an item in an environment. For example, conventional systems may indicate a general area (e.g., a room) in an environment where an item may be located, but are unable to specifically point out the precise position of the item. Further, as each item requires a tracking device, items that do not have a tracking device cannot be tracked or located through existing systems. Also, conventional systems for tracking and assisting users in locating items in an environment may generate a path using a two-dimensional (2D) map of the environment or display the general direction of or distance to an item. These approaches are limited in their ability to account for obstacles or other features of the environment, which may impact a user's ability to reach and/or identify the item.

SUMMARY

(2) Embodiments of the present disclosure relate to maintaining memory of objects using a 3D reconstruction of an environment. Systems and methods are disclosed that may be used to track and locate items within a 3D representation of an environment, provide alerts regarding the items, provide recommendations for storage locations of the items, and plan routes to the items.

(3) In contrast to conventional systems, such as those described above, a 3D representation of an environment may be generated from sensor data, with objects being detected in the environment using the sensor data and stored as items that can be tracked and located within the 3D representation. The 3D representation of the environment and item information may be used to determine (e.g., identify or predict) a location of an item within the 3D representation and/or recommend a storage location for the item within the 3D representation. Using a determined location, one or more routes to the location through the 3D representation may be determined. Data corresponding to a determined route may be provided to a user and/or device. User preferences, permissions, roles, feedback, historical item data, and/or other data associated with a user may be used to further enhance various aspects of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present systems and methods for maintaining memory of objects using a 3D reconstruction of an environment are described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1 is an example system diagram of an item tracking and locating system, in accordance with at least some embodiments of the present disclosure;

(3) FIG. 2 is an illustration of an example flow diagram for generating and/or updating one or more aspects for a 3D model of an environment, in accordance with some embodiments of the present disclosure;

(4) FIG. 3A is an illustration of an example user interface for indicating a location of an item in an environment, in accordance with some embodiments of the present disclosure;

(5) FIG. 3B is an illustration of an example client device presenting a user interface including an item-based alert and an item-based recommendation, in accordance with some embodiments of the present disclosure;

(6) FIG. 4 is a flow diagram showing a method a server may perform for tracking and locating

items based on 3D reconstruction, in accordance with some embodiments of the present disclosure;

(7) FIG. 5 is a flow diagram showing a method a client device may perform for tracking and locating items using a 3D reconstruction of an environment, in accordance with some embodiments of the present disclosure;

(8) FIG. 6 is a flow diagram showing a method for tracking and locating items using a 3D representation of an environment, in accordance with some embodiments of the present disclosure;

(9) FIG. 7 is a block diagram of an example computing device suitable for use in implementing some embodiments of the present disclosure; and

(10) FIG. 8 is a block diagram of an example data center suitable for use in implementing some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) Systems and methods are disclosed related to maintaining a persistent memory of object state and position information using a 3D reconstruction of an environment. Disclosed approaches may be used to track and locate items corresponding to the objects within a 3D representation of an environment, provide alerts regarding the items, provide recommendations for storage locations of the items, and plan routes to the items.

(12) In contrast to conventional systems, such as those described above, a 3D representation of an environment may be generated from sensor data, with objects being detected in the environment using the sensor data and stored as item representations that can be tracked and located within the 3D representation. In at least one embodiment, one or more videos of the environment may be captured using a client device, such as a mobile device. The 3D representation of the environment may be generated and/or updated using a 3D reconstruction of at least some of the environment generated, for example, using photogrammetry. One or more objects may be detected and segmented from the 3D reconstruction and associated with one or more items (e.g., to determine a 3D model for an object and corresponding item). One or more attributes of an item may be determined based on a corresponding object(s), such as those indicating one or more of a size of the object, a shape of the object, a pose of the object, a surface normal of the object, accessibility of the object, a location of the object, an orientation of the object, one or more surface normals of the object, a class or category of the object, or an affordance(s) of the object.

(13) In at least one embodiment, the 3D representation of the environment and item information may be used to determine (e.g., identify or predict) a location of an item within the 3D representation and/or recommend a storage location for the item within the 3D representation. The determination may be based on item attributes and/or 3D geometry of the environment and objects in the environment, which may allow for more accuracy in predictions and/or determinations. Using a determined location, disclosed approaches may plan one or more routes to the location through the 3D representation while accounting for 3D geometry of the environment, 3D item location, and/or attributes of the user and/or device or machine that is to navigate a route (e.g., physical attributes and/or capabilities). Data corresponding to a determined route may be provided to a user and/or device, for example, within an interface displaying at least some of the 3D representation. User preferences, permissions, roles, feedback, historical item data, and/or other data associated with a user may be used to further enhance the accuracy of determinations, provide alerts to seek out items, provide personalized route planning to items, and/or provide security, privacy, and personalization for items and/or 3D representations of environments.

(14) The systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, these purposes may include systems or applications for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, autonomous or semi-autonomous machine applications, deep learning, environment simulation, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, digital twin systems, cloud computing and/or any

other suitable applications.

(15) Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems implemented using an edge device, systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems for generating or maintaining digital twin representations of physical objects, systems implemented at least partially using cloud computing resources, and/or other types of systems.

(16) FIG. 1 is an example system diagram of an item tracking and locating system **100** (also referred to as “system **100**”), in accordance with at least some embodiments of the present disclosure. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.

(17) The system **100** may include, among other features, one or more client devices **102**, one or more servers **104**, and one or more network(s) **108**. Although one client device **102** is illustrated in FIG. 1, this is not intended to be limiting. In examples, there may be any number of client devices **102**. The system **100** (and the components and/or features thereof) may be implemented using one or more computing devices, such as the computing device **700** of FIG. 7, described in more detail below.

(18) The client device(s) **102** may include an application **110**, an input device(s) **112**, a communication interface **114A**, a display **116**, and a sensor(s) **118**. Although some components and/or features of the client device(s) **102** are illustrated in FIG. 1, this is not intended to be limiting. For example, the client device(s) **102** may comprise additional or alternative components, such as those described below with respect to the computing device **700** of FIG. 7.

(19) The server(s) **104** may include, among other things, a communication interface **114B**, an environment manager **120**, an environment modeler **122**, an environment renderer **124**, an alert manager **126**, a recommendation manager **128**, a route manager **130**, a user profile manager **132**, a query engine **138**, and a data store(s) **146**. Although some components and/or features of the server(s) **104** are illustrated in FIG. 1, this is not intended to be limiting. For example, the server(s) **104** may comprise additional or alternative components, such as those described below with respect to the computing device **700** of FIG. 7 and/or the data center **800** of FIG. 8.

(20) As an overview, the communication interface **114A** of a client device **102** may be configured to manage internal and/or external communications for the client device **102**, such as communications between the client device **102** and the server **104** over the network **108**. Similarly, the communication interface **114B** of the server **104** may be configured to manage internal and/or external communications for the server(s) **104**, such as communications between the client device **102** and the server **104** over the network **108**. The environment manager **120** of the server **104** may be configured to manage one or more 3D virtual environments (e.g., that are based on geometry of real-world environments). To this effect, the environment manager **120** may use the environment modeler **122** configured to generate and/or update one or more 3D models of at least one of the one

or more 3D virtual environments and/or portions thereof, such as one or more objects thereof. The one or more 3D models may correspond to at least a portion of model data **134** stored in the data store(s) **146**. The environment manager **120** may further use the environment modeler **122** configured to generate and/or update one or more items corresponding to the one or more objects and/or one or more attributes thereof. The one or more items and attributes may correspond to at least a portion of item data **136** stored in the data store(s) **146**.

(21) The application(s) **110** of the client device(s) **102** may be configured to provide one or more user interfaces, which may be used to interact with the server(s) **104**, such as a user interface **140A** and/or a user interface **140B** (which may be referred to as “user interfaces **140**”). As indicated in FIG. **1**, the display **116** may be configured to display at least a portion of the user interface(s) **140**. The input devices **112** may be configured to receive user inputs, for example, to the one or more user interfaces **140**. The sensor(s) **118** may be configured to generate sensor data, such as image data. In at least one embodiment, the client device **102** may provide the sensor data to the server(s) **104**, via the communication interface **114A**, for use by the environment modeler **122** in generating and/or updating the one or more 3D models or other aspects of a 3D virtual environment.

(22) The environment renderer **124** of the server **104** may be configured to render one or more aspects of the one or more 3D virtual environments, for example, using the one or more 3D models and/or attributes provided by the environment modeler **122**. In at least one embodiment, the environment renderer **124** may render one or more images and/or frames, such as the image **142**, using 3D graphics rendering, which may include light transport simulation (e.g., ray-tracing, path tracing, ray marching, light reservoir sampling, etc.). In at least one embodiment, the server **104** may provide data corresponding to the rendering to one or more of the client device(s) **102**, via the communication interface **114B**, for use by the application(s) **110** in presenting and/or updating content displayed in the one or more user interfaces **140**.

(23) The alert manager **126** may be configured to generate and provide one or more alerts to the client device(s) **102**, which may be included in the content displayed in the one or more user interfaces **140**. The recommendation manager **128** may be configured to generate and provide one or more recommendations to the client device(s) **102**, which may be included in the content displayed in the one or more user interfaces **140**. The route manager **130** may be configured to generate and provide one or more routes to the client device(s) **102**, which may be included in the content display in the one or more user interfaces **140**. The user profile manager **132** may be configured to record, store, verify, and/or update one or more user preferences, permissions, roles and/or other data associated with a user, which may be used by any of the various components of the server **104** to provide personalization and/or security to the system **100**. The query engine **138** may be configured to process queries regarding the items stored in the data store **146**, such as for one or more locations of an item. In any embodiment, such as where personal, confidential, and/or provide information is available, strict security and data privacy technologies and procedures may be employed to restrict unauthorized access to information stored. Example non-limiting technologies include encryption and identity verification (e.g., via multi-factor authentication, and/or password/pin code). For example, one or more 3D representations of one or more environments may be encrypted, in addition to any user profile data and item data, and access to the representations or user profile and item data may be restricted to only users with verified identities.

(24) Components of the system **100** may communicate over network(s) **108**. The network(s) may include a wide area network (WAN) (e.g., the Internet, a public switched telephone network (PSTN), etc.), a local area network (LAN) (e.g., Wi-Fi, ZigBee, Z-Wave, Bluetooth, Bluetooth Low Energy (BLE), Ethernet, etc.), a low-power wide-area network (LPWAN) (e.g., LoRaWAN, Sigfox, etc.), a global navigation satellite system (GNSS) network (e.g., the Global Positioning System (GPS)), and/or another network type. In any example, each of the components of the system **100** may communicate with one or more of the other components via one or more of the networks **108**.

(25) The display **116** may include any type of display capable of displaying a user interface (e.g., a light-emitting diode display (LED), an organic LED display (OLED), a liquid crystal display (LCD), an active matrix OLED display (AMOLED), a quantum dot display (QDD), a plasma display, an LED/LCD display, and/or another type of display). In some examples, the display **116** may include more than one display (e.g., a dual-monitor display). In some examples, the display **116** includes a touch-screen display, such as a touch-screen of a smart phone, tablet computer, laptop computer, or the like, where the touch-screen corresponds to at least one of the input device(s) **112** of a client device **102**.

(26) The client devices **102** may include one or more smart phones, laptop computers, tablet computers, desktop computers, robots, autonomous machines, autonomous vehicles, wearable devices, edge devices, game consoles, virtual reality systems (e.g., a headset, a computer, a game console, remote(s), controller(s), and/or other components), streaming devices, smart-home devices that may include an intelligent personal assistant, and/or other types of devices capable of supporting at least some of the functionality described herein.

(27) In some examples, the client device(s) **102** may include one or more portions of the environment renderer **124**, and may render the one or more aspects of the one or more 3D environments (e.g., using the application **110**), while in other examples, the client device(s) **102** may receive display data (e.g., encoded video data) and use the display data to display the one or more portions of the one or more user interfaces **140** on the display **116**, such as an image **142** (e.g., rendered using 3D graphics processing) of a 3D virtual environment **144** (which may be a 3D representation of the environment **150** and referred to as such). In some examples, a first client device **102** may render the one or more aspects while a second client device **102** may receive the display data and display the one or more aspects using the display data. Thus, any number of client devices **102** may be used to carry out one or more portions of disclosed approaches and any of the client devices **102** may have different (e.g., fewer or additional) components, as needed or required to carry out those portions. In one or more embodiments, any of the various components shown in FIG. **1** may be embodied, at least in part, on any number or combination of client devices **102** and/or servers **104**.

(28) The application **110** may be referred to as a single application for simplicity, but its functionality can be embodied by one or more applications in practice. Other devices can include one or more applications similar to the application **110**. The application(s) **110** may include one or more mobile applications, computer applications, console applications, video game applications, computer programs, services, processes, and/or another type of application. It is contemplated herein that “application” be interpreted broadly. As an example, an application **110** can be installed on an operating system of a client device **102**. As another example, an application **110** can include a service integrated into an operating system of a client device **102**. In various implementations, an application **110** can be on a client device **102**, as shown with respect to the application **110**, but can at least partially be hosted server-side (e.g., on a server **104**). In some cases, the application **110** comprises a web browser interfacing with a server **104** via a web site.

(29) The input device(s) **112** may include any type of device(s) capable of providing user inputs to the user interfaces **140**. The input device(s) **112** may include a keyboard, a mouse, a microphone(s), a touch-screen display, a controller(s), a remote(s), a headset (e.g., sensors of a virtual reality headset), and/or other types of input devices.

(30) The sensor(s) **118** may include at least one sensor capable of generating sensor data representative of at least some aspect of an environment, such as an environment **150**. For example, the sensor(s) **118** may generate sensor data **148**, which the environment modeler **122** may use to generate and/or update one or more models corresponding to at least one of the one or more 3D virtual environments, such as a 3D virtual environment **144** (e.g., a 3D representation of the environment **150**). The sensor(s) **118** may include any combination of an image sensor(s), a global navigation satellite systems (GNSS) sensor(s) (e.g., Global Positioning System (GPS) sensor(s)), a

RADAR sensor(s), an ultrasonic sensor(s), LIDAR sensor(s), an inertial measurement unit (IMU) sensor(s) (e.g., accelerometer(s), gyroscope(s), magnetic compass(es), magnetometer(s), etc.), microphone(s), stereo camera(s), wide-view camera(s) (e.g., fisheye cameras), an infrared camera(s), a surround camera(s) (e.g., 360 degree cameras), a long-range and/or mid-range camera(s), a speed sensor(s) (e.g., for measuring the speed of the client device **102** and/or a robot), a vibration sensor(s), a steering sensor(s), a brake sensor(s) (e.g., as part of the brake sensor system), and/or other sensor types. The environment **150** may include many types of environments, such as an indoor environment, a warehouse, a store, a home, a manufacturing plant, a factory, a private property, a public property, etc.

(31) In some examples, the sensor data **148** may be generated by forward-facing and/or side-facing cameras, such as a wide-view camera(s), a fisheye or surround-view camera(s), a stereo camera(s), and/or a long-range or mid-range camera(s). In some examples, more than one camera or other sensor may be used to incorporate multiple fields of view.

(32) While the sensor(s) **118** is shown as being included in the client device **102**, this need not be the case. For example, the sensor(s) **118** used to generate one or more portions of the sensor data **148** may be included on any number of devices, such as one or more client devices **102** and/or other devices. In at least one embodiment, the sensor(s) **118** are included, at least in part, in one or more robots, such as autonomous mobile robots (AMRs), one or more security cameras, one or more mobile devices (e.g., a smart phone), etc.

(33) One or more embodiments may include multiple robots for performing one or more portions of item retrieval or data collection. By way of example, and not limitation, one or more robots may travel vertically and/or one or more robots may travel horizontally (e.g., a vertical or horizontal tracks). Each robot may have a specific set of grasping tools that allow them to hold and transport different types of items depending on different form factors of the items. Additionally or alternatively, different robots or other devices may provide different viewpoints for data collection and/or environmental reconstruction.

(34) As described herein, the environment modeler **122** may be configured to generate and/or update one or more 3D models of a 3D virtual environment and/or portions thereof and generate and/or update one or more items corresponding to one or more objects and/or one or more attributes thereof. Referring now to FIG. 2, FIG. 2 is an illustration of an example process **200** that may be used to generate and/or update one or more aspects of one or more 3D virtual environments, in accordance with some embodiments of the present disclosure.

(35) The process **200** may include image generation **220** to generate one or more images **210** that depict one or more portions of the environment **150**. For example, the images **210** may include images of the environment **150** from different perspectives. The images **210** may include RGD and/or RGB-D images. In at least one embodiment, the images **210** include one or more images from a sequence of images capturing the environment **150**. For example, responsive to user input from the input devices **112**, a client device **102** may record a video of the environment **150** using the sensor(s) **118**, and the video may correspond to at least some of the sensor data **148** (e.g., the client device **102** may be a smart phone that records video as a user walks through the environment **150**). Additionally or alternatively, the client device **102** may be or be included on a robot that records a video of the environment **150** as it traverses and/or moves in the environment **150**. In further examples, multiple videos from multiple wall-mounted security cameras could be used. The image generation **220** may be performed using the user interfaces **140** and/or the user interfaces **140** may be used to facilitate import of the images **210**, which may be generated using one or more other user interfaces. In some examples (e.g., where a robot captures the images **210**), no user interface **140** may be employed or a user interface **140** may be operated from another client device **102**.

(36) The process **200** may also include 3D reconstruction **212** to generate and/or update one or more portions of one or more 3D models of the environment **150**. The 3D reconstruction **212** may

be performed by the environment modeler **122** and may use any of a variety of 3D reconstruction techniques to generate and/or update a 3D representation **250** (or reconstruction) of the environment **150**. In at least one embodiment, the environment modeler **122** generates the 3D representation **250** using photogrammetry. In at least one embodiment, the environment modeler **122** generates the 3D representation **250** using one or more GPUs and/or cloud processing.

(37) The process **200** may further include object segmentation **224** to generate and/or update one or more portions of one or more 3D models and/or attributes of one or more items stored in association with the 3D virtual environment **144** (item metadata). The object segmentation **224** may be performed by the environment modeler **122** and may be used to recognize and extract one or more portions of objects, such as objects **240** and **242** from the 3D representation **250** (e.g., a point cloud). In at least one embodiment, the object segmentation **224** may use the environment modeler **122** to generate and/or retrieve 3D models for objects depicted in the images **210**. For example, an approach may be used that employs surface normal predictions, along with appearance cues, to retrieve 3D models from an object model library **246** (e.g., computer-aided design models). The 3D models may be used to generate and/or update 3D models for the one or more objects. In one or more embodiments, one or more machine learning models (MLMs), such as convolutional neural networks (CNNs), may be used for surface normal prediction. The surface normals computed from the images **210** may be used to predict an object pose and the pose may be used to match the object to a model, as indicated in FIG. **2** with respect to the objects **240** and **242**.

(38) For example, and without limitation, an MLM(s) described herein may include any type of machine learning model, such as a machine learning model(s) using linear regression, logistic regression, decision trees, support vector machines (SVM), Naïve Bayes, k-nearest neighbor (Knn), K means clustering, random forest, dimensionality reduction algorithms, gradient boosting algorithms, neural networks (e.g., auto-encoders, convolutional, recurrent, perceptrons, Long/Short Term Memory (LSTM), Hopfield, Boltzmann, deep belief, deconvolutional, generative adversarial, liquid state machine, etc.), and/or other types of machine learning models. In various examples, an MLM may include one or more convolutional neural networks.

(39) Additionally or alternatively, the environment modeler **122** may determine one or more portions of a 3D model for an object using mesh inputs. For example, example, a mesh model may be generated for an object where the environment modeler **122** determines it is unable to match the object with a model from the library.

(40) In at least one embodiment, an item may be updated based at least on determining an observed object corresponds to a previously observed object associated with the item. For example, one or more MLMs, such as one or more twin neural networks, may be used to detect a previously observed object(s). A twin neural network may use the same weights while working in tandem on multiple different input vectors to compute comparable output vectors. One or more of the output vectors (e.g., for the previously observed object) may be precomputed and used as a baseline against which one or more other output vectors are compared (e.g., for the observed object being analyzed). In at least one embodiment, one or more of the twin neural networks may be trained using triplet loss with margin alpha. One-shot learning may be used at inference time. In various examples, one or more of the twin neural networks may be used to personalize object and/or item recognition at any suitable time described herein, which an item or object may be detected (e.g., to confirm storage at a location, to determine an item that is a subject of a query, to determine the item is moving or was moved and the location is to be updated, etc.). In various embodiments, the sensors **118** from one or more client devices **102** and/or other devices may form a sensor network, such as a wireless sensor network providing multiple data collection sources spatially and/or temporally (e.g., sets of LIDAR data, sets of image data, etc.). In order to consolidate the data sources, such as to account for potentially redundant data, the environment modeler **122** may use various approaches such as twin neural networks and self-supervised networks.

(41) In at least one embodiment, the environment modeler **122** may use the process **200** and/or

other processes and analysis to generate and/or update one or more items corresponding to the one or more objects extracted using the images **210**. For example, the one or more items may be stored in the item data **136** in the data store(s) **146**. Each item may include an item identifier stored in association with one or more of the 3D models (e.g., determined using the process **200**) and may correspond to one or more of the objects. Further, each item or item ID may be stored in association with one or more attributes. The one or more attributes may be determined using the process **100** and/or additional analysis or processing of the sensor data **148**. In various examples, one or more of the attributes may be associated with geometry of the item and/or corresponding object(s).

Examples of attributes may include one or more attributes indicating one or more of a size of the one or more objects, a shape of the one or more objects, a pose of the one or more objects, a surface normal of the one or more objects, accessibility of the one or more objects (e.g., whether the item is reachable by fork lift, whether the item is reachable by a person at ground level or using a ladder, etc.), or an affordance(s) of the one or more objects.

(42) The user profile manager **132** may be configured to record, store, verify, and/or update one or more user preferences, permissions, roles and/or other data associated with a user, which may be used by any of the various components of the server **104** to provide personalization and/or security to the system **100**. For example, the user data may be used to limit access to items by user and/or role. Thus, while an environment may include multiple items for multiple users, only users with permissions to search for an item may be granted access to a predicted or known location of the item. For example, one or more users may be assigned as owners of an item with the ability to grant and/or revoke permissions and/or access for other users regarding the item. Thus, when a user searches for a generic keyword like “keys,” the results returned and/or displayed may only be for the user's keys and/or keys the user is authorized to access.

(43) Additionally or alternatively, the user data may be used to limit access to 3D environments and/or portions thereof by user and/or role. Thus, while multiple environments may be stored by the server **104** or an environment may include multiple areas, only authorized users with sufficient/requisite permissions or rights to view and/or search for an item in that area or granted access to information of an environment. For example, one or more users may be assigned as owners of a 3D environment or area with the ability to grant and/or revoke permissions and/or access for other users regarding the 3D environment or portion thereof. In at least one embodiment, a user may be assigned as an owner of an area and/or environment based at least on using the client device **102** to generate or create the area and/or 3D environment (e.g., to capture or provide the sensor data used to generate the environment).

(44) In at least one embodiment, the attributes may correspond to one or more locations or positions of the item, for example, relative to the 3D virtual environment **144** (e.g., coordinates, location label, and/or pose, for example, if observed). By way of example and not limitation, the environment modeler **122** may determine a location or position of the object(s) corresponding to the item, and record the location or position with respect to the item. Suitable approaches for determining the location or position may include localization with respect to the 3D reconstruction **212** and/or the environment **150**. Localization may include odometry, landmark localization, pose estimation, simultaneous localization and mapping (SLAM), and the like.

(45) In at least one embodiment, the environment modeler **122** may store multiple locations or positions (e.g., one or more coordinates in the 3D virtual environment and/or real environment) for a single item. One or more of the locations or positions may be marked as a most recent location or position, for example, based at least on the environment modeler **122** most recently observing the item at the location or position (e.g., using the process **200**, localization, etc.) and/or based at least on data from one or more users indicating or identifying the location or position. For example, when the environment modeler **122** determines an item is at a new location or position, the environment modeler **122** may update the most recent location or position stored in the item data **136**. One or more previous locations may be retained and/or analyzed for determining

recommendations and/or probabilities with respect to the location(s) or position(s) of the item, as described herein.

(46) In at least one embodiment, a 3D virtual environment may be updated over time. For example, the process **200** and/or the 3D reconstruction **222** for the 3D virtual environment **144** may be performed at times t , $t+1$, $t+2$, and so forth to generate corresponding snapshots of the environment **150**. In at least one embodiment, the snapshots may store differences and/or changes between the snapshots so as to reduce storage. For example, if only an item location changes between t and $t+1$, the snapshot for $t+1$ may only store the new location for the item (e.g., and pointers or other stored values changed for one or more other snapshots). In at least one embodiment, an update to the 3D virtual environment **144** may be initiated by a user and/or client device **102**. For example, the user may provide, via the input device **112**, user input to initiate an update to at least a portion of the 3D virtual environment **144** (e.g., using a user interface **140**). In at least one embodiment, continuous sensor data recording and/or updating of the 3D virtual environment **144** may be avoided. For example, a user may choose to provide video when an object has been added to the environment **150** or an object has been moved. The video may change the item data **136** and/or the model data **134** while maintaining user preferences.

(47) Referring now to FIG. 3A, FIG. 3A is an illustration of an example user interface **340** for indicating a location of an item in an environment, in accordance with some embodiments of the present disclosure. The user interface **340** may be included in the user interfaces **140** of FIG. 1. The user interface **340** includes a location indicator **350**, which may be provided by user input to the input devices **112**. For example, the location indicator **350** may include a pin or other marker dropped into a depiction of the environment **150**, such as an image of the environment **150**, the 3D virtual environment **144** (e.g., using augmented reality), or a corresponding map. The location indicator **350** may identify or indicate the location or position of the item in the 3D virtual environment **144** and/or the environment **150** to the environment modeler **122**, which may be recorded by the environment modeler **122**. While the location indicator **350** is shown as an example, other examples of location indicators may be used that identify or otherwise indicate a location or position, such as verbal indicators, descriptive textual indicators, etc.

(48) In one or more embodiments, the recommendation manager **128** may recommend one or more locations or positions for an item. In such examples, the location indicator may correspond to a selection of a location or a position of the one or more locations or positions (e.g., from a list of locations or positions). Additionally or alternatively, the data indicating the location or position may include location data associated with the user and/or client device **102**, such as sensor data generated using the sensor(s) **118** and/or other sensors in the environment **150**. For example, the environment modeler **122** may analyze the data (e.g., image analysis, motion analysis, gyroscope-based analysis, etc.) to determine the item was or is being placed in the recommended location.

(49) The query engine **138** may be configured to process queries regarding the items stored in the data store **146**, such as for one or more locations of an item. The queries may include user submitted queries, queries submitted from a client device **102**, and/or system submitted queries. In at least one embodiment, the query engine **138** may locate an item based at least on searching the 3D virtual environment(s) **144** in the data store(s) **146** for the item. In at least one embodiment, the query engine **138** may be incorporated, at least in part, into a personal assistant. The personal assistant may be accessed, for example, using a client device **102**. In one or more embodiments, the query engine **138** may be optimized for rapid retrieval, allowing for the system **100** to be more performant, personalized, and have reduced latency while enforcing privacy constraints, for example, by running on the user's device itself. In one or more embodiments, searching may be executed using a lightweight process, e.g., by operating on pre-computed metadata which may be included in the item data **136**.

(50) A query may include an image-based and/or text-based query. For example, a query may include data representing one or more images. The user interface **140A** is an example of an

interface which may be used to provide at least one of the images of a query. For example, user input via the input device **112** may select an image **152** for a query using the user interface **140A**. In one or more embodiments, the image **152** may be selected from a list of images, with each image corresponding to an item in the data store **146**. Thus, the image **152** may identify and/or be stored in association with the item(s) for the query. In further examples, the image(s) **152** may be a photograph or other image, which may not be stored in association with an item in the data store **146**. In such examples, in processing the query, the query engine **138** may analyze the one or more images **152** to determine one or more features of the image (e.g., using one or more MLMs). The query engine **138** may then select and/or identify one or more items from the one or more items in the data store **146** for the query based at least on the one or more features and the one or more attributes of the item(s). For example, the user may take a photo and/or provide a live stream of video of an object for a query.

(51) The user interface **140A** also provides an example of an interface which may be used to provide text and/or audio for a query, such as natural language text. For example, user input via the input device **112** may provide a text-based query **154** (e.g., a natural language query via text and/or audio input) using the user interface **140A**. In one or more embodiments, the query engine **138** may use the text-based query **154** to identify and/or select the item(s) for the query. For example, in processing the query, the query engine **138** may analyze the text of the query to determine one or more features of the text (e.g., using one or more natural language processing techniques). The query engine **138** may then select and/or identify one or more items from the one or more items in the data store **146** for the query based at least on the one or more features and the one or more attributes of the item. Examples of text-based queries include “Where is the fork lift?,” “Office Keys,” “Find pallet #7584,” etc. In addition to or alternatively from image-based and/or text-based queries, other forms of user input may be used to define a query. For example, a user may provide user input via an input device **112** that modifies one or more object properties, object classes, and/or item data **136** (e.g., metadata) to define a query. Using such human-in-the-loop inputs may improve query quality.

(52) Examples of the features include one or more of text, keywords, object categories, object classes, query submission time, semantic information regarding one or more portions or view of an environment corresponding to the query (e.g., depicted in an image(s)), one or more object sizes and/or dimensions, one or more confidence scores associated with one or more other features and/or inferences made using an MLM, image color information, image luminance information, detected object location with respect to one or more other objects in the environment, a user ID and/or identifier associated with the query, features indicating one or more of a size of one or more objects, a shape of one or more objects, a pose of one or more objects, a surface normal of one or more objects, an affordance(s) of one or more objects, feature embeddings, etc.

(53) In at least one embodiment, the query engine **138** may, responsive to a query, search the item metadata and/or attributes stored in the item data **136** (e.g., an index of items) for one or more matching items (e.g. to the query features). By way of example and not limitation, the query engine **138** may extract one or more keywords and/or text features from the query for a comparison with the item data **136** for the search. For example, the query engine **138** may search for an exact text match of an item. In some cases, the search may identify multiple matches and/or similar matches. For example, the query engine **138** may assign similarity or confidence scores to an item that quantifies a likelihood the query corresponds to the item (e.g., based on similarities in item metadata to query features).

(54) By way of example and not limitation, the item data **136** may include an item of each key of twenty keys. If a user searches for keys, based on a time and/or other inputs, as well as the user, the query engine **138** can attempt to narrow down the twenty keys to one or a subset of keys that may be the subject of the query. If there is a specific time a user profile maintained by the user profile manager **132** indicates that the user typically takes its office keys, that personalization may be

factored into the narrowing process, which may result in at least an item representing the office keys being selected (e.g., due to having a highest confidence score, due to a confidence score exceeding a threshold, etc.).

(55) Having determined an item corresponding to a query, the query engine **138** may search for the item in a 3D virtual environment so as to provide one or more potential locations or positions in response to the query. Various factors may be used to select, filter out, and/or rank or score one or more locations or positions for an item. For example, in at least one embodiment, the query engine **138** may provide a location or position for the item based at least on the location or position being marked as a most recent location or position for the item. Additionally or alternatively, the query engine **138** may provide a location or position for the item based at least on a time (e.g., storage time) corresponding to the location or position associated with the item. For example, the longer the time the less likely the location or position may be provided and vice versa. Additionally or alternatively, the query engine **138** may provide a location or position for the item based at least on a frequency (e.g., storage frequency) corresponding to the location or position associated with the item. For example, the lower the frequency the less likely the location or position may be provided, and vice versa. Additionally or alternatively, the query engine **138** may provide a location or position for the item based at least on one or more affordances and/or other attributes of the item and/or an item corresponding to the location. For example, one or more locations or positions may be selected or scored higher based at least on the query engine **138** determining compatibility with one or more of the affordances and/or other item attributes and vice versa. This may include determining an item and corresponding location or position are geometrically compatible, such as that the item will physically fit within a region of the 3D reconstruction corresponding the location or position, the item can sit at rest at the location or position (e.g., won't fall or slip), the geometry of the location or position allows for an affordance or use of the item, etc.

(56) By way of example, where the location or position is a drawer and the item is a set of keys, the attributes may indicate the drawer can be used for storing keys and/or the keys can be stored in a drawer. As described herein, the affordances may be determined based at least on the geometry of the object(s) corresponding to the item (e.g., using a 3D model to determine whether the item would fit in a drawer) and/or a category or class assigned to the item. As another example, if the item is a set of keys that are for driving a vehicle, the affordance may indicate the keys can be located in a vehicle.

(57) The one or more locations or positions may be provided to the user, for example, in a user interface **140** responsive to the query. Providing a location or position may include presenting and/or displaying one or more indications of the location or position. For example, the location or position may be provided using a location identifier or indicator, which may be presented on a client device **102** audibly and/or visually. In at least one embodiment, a location indicator may include a render of the one or more locations or positions in the 3D virtual environment displayed on the client device **102** (e.g., rendered by the environment renderer **124**). Additionally or alternatively, a name of the location or a description of the position in the environment (e.g., adjacent or proximate to one or more other objects, relative to one or more other objects, etc.) may be displayed on the client device **102**.

(58) In one or more embodiments, the route manager **130** may plan one or more routes to at least a location of the one or more locations through the 3D representation of the environment. For example, based at least on the query engine **138** selecting a location or position for a response to a query and/or a user selection of the location or position (e.g., in the presentation to the user responsive to the query), the route manager **130** may plan a route to the location and/or display or present the route to the user (e.g., in a user interface **140**). Planning a route to a location may be based at least on one or more attributes of the user and/or client device **102** (e.g., robot) that is to traverse the route. The one or more attributes may be determined from user profile data managed by the user profile manager **132**. Examples include one or more attributes indicating one or more of

a location or positions corresponding to the user and/or the client device **102** (e.g., a current location, orientation, pose, etc. associated with the query), a size or other dimension(s) of the user and/or client device **102**, a shape of the user and/or client device **102**, a pose of the user and/or client device **102**, physical capabilities of the user and/or client device **102**, etc.

(59) In at least one embodiment, one or more paths for one or more routes may be determined and/or selected based at least on the geometry of the 3D virtual environment and the one or more attributes. For example, a route **156** is indicated in FIG. **1**, which may account for an object **158** (or obstacle) in the environment **150** as indicated by the 3D virtual environment **144**. For example, for a robot, a path may go under a table based on the height of the robot being less than the table. However, for a person, the route may avoid or be less likely to go under the table based at least on the height of the user being greater than the table, and/or the physical capabilities associated with the user (e.g., based on the user having crutches). Using disclosed approaches, path suggestions or selections may be personalized to the user and/or device. For example, different robots in a warehouse may receive different path suggestions by accounting for different dimensions and/or maneuvering capabilities of the robots. As a further example, different people may receive different path suggestions by accounting for disabilities.

(60) In at least one embodiment, one or more cost maps may be used for route planning. In some embodiments, route planning may include computing one or more distance fields (corresponding to distances to objects and/or surfaces in the environment), and from the one or more distance fields (e.g., a Euclidean Signed Distance Field) the cost map may be derived. The cost map may correspond to a volumetric reconstruction of the physical environment, for example, determined using the process **200**. The cost map may be used for collision avoidance and/or path planning. For example, the cost map may be used by one or more downstream navigation components used for path planning, object avoidance, localization, and/or other operations for controlling the client device **102** and/or facilitating navigation of the user through the physical environment.

(61) By way of example, the route **156** may be generated based at least on the location **160** associated the user and/or client device **102** and/or the location **162** associated with the item. The route **156** may be multi-dimensional and represented using any suitable approach, such as using one or more trajectories, points, coordinates, and/or functions. A determined route or path may include vertical and/or horizontal components. In a data center, for example, a route for a robot may include horizontal and vertical segments. In one or more embodiments, a route may be determined in 3D, for example, traversing ramps and other objects or obstacles in the environment. For a machine capable of aerial travel, a route may include one or more aerial segments. In at least one embodiment, the route **156** may be generated and/or selected based at least on the route manager **130** determining the route **156** is the shortest route that satisfies one or more hard or soft constraints associated with the user and/or client device **102**. While the route **156** is shown using a path through an environment, the route **156** may be presented using audible directions, textual directions, step-by-step directions, etc. In one or more embodiments, the presentation of the route **156** may point to or otherwise indicate a particular location corresponding to the item. In at least one embodiment, the environment renderer **124** may present the item in a visual distinct manner in the user interface **140**, such as using highlighting, a changed color(s) etc. This may assist a user in located the item in the environment.

(62) As described herein, the recommendation manager **128** may recommend one or more locations or positions for storage of an item(s). The one or more recommended locations or positions for an item may be selected and/or determined using similar approaches and/or factors as described with respect to the query engine **138** determining one or more locations or positions for an item. Thus, for example, one or more locations or positions may be recommended for storage based at least on the recommendation manager **128** determining the item will physically fit in the locations and/or at the positions (e.g., one or both using the 3D representation **250**), that the item was previously stored in the locations or positions, and/or that the locations or positions are preferred locations or

positions based on storage types specified by the user, amongst other factors. Referring now to FIG. 3B, FIG. 3B is an illustration of an example client device **102** presenting a user interface (e.g., corresponding to a user interface **140**) including an item-based alert **304** and an item-based recommendation **306**, in accordance with some embodiments of the present disclosure.

(63) While the item-based recommendation **306** and the item-based alert **304** are shown as being presented on a display **116** of the client device **102**, one or more item-based recommendations and/or item-based alerts may be presented using any suitable modality, including modality described herein with respect to results from the query engine **138**. In at least one embodiment, a recommendation may be selected by a user and/or client device (e.g., via the input device **112**). For example, a user may select a graphical interface element corresponding to the item-based recommendation **306** to select the corresponding location. As a further example, the client device **102** and/or may implicitly select the corresponding location, for example, by moving to or toward the location, storing the item in the location, etc. In at least one embodiment, the user profile manager **132** may, based at least on determining the location or position was selected, update corresponding user profile data, for example, to record the location or position as a current storage location or position and/or to update data indicating a frequency of the item being stored by the user at the location or position. In at least one embodiment, the storage location or position may be determined, or confirmed, based at least on analyzing sensor data from the sensors **118**, such as image data, motion data, and/or location data for determining or predicting confirmation of storage (e.g., visual confirmation).

(64) In at least one embodiment, the route manager **130** may plan one or more routes to at least a location of the one or more locations recommended by the recommendation manager **128** through the 3D representation of the environment. For example, a route may be determined and/or presented similar to what is described herein for the route **156**, but using a recommended location (which may include pose, orientation, height, 3D coordinates, adjacent or proximate objects, position relative to other objects, etc., that may be visually indicated to the user).

(65) As described herein, the alert manager **126** may be configured to generate and provide one or more alerts to the client device(s) **102**, such as the item-based alert **304**. In at least one embodiment, the item-based alert **304** may be presented based at least on determining and/or predicting one or more storage and/or retrieval times for one or more items based at least on historical storage and/or retrieval data, which may be included in the user profile data maintained by the user profile manager **132**. For example, the historical data may correspond to any number of prior storage and/or retrieval times and/or locations for one or more items and/or similar items for one or more users (e.g., the historical data may be personalized to the user and/or be based on aggregated user data). Any suitable approach may be used to determine an item-based alert, such as event prediction and/or anticipation techniques (e.g., statistical, MLM, etc.), where the event may include storage and/or retrieval of the item, any item, and/or an item class or type. Additionally or alternatively, one or more alerts may be scheduled by the user and/or client device **102**. For example, a reminder may be provided every day and/or based at least on determining one or more user and/or system configured trigger criteria are satisfied. In one or more embodiments, the criteria may be based at least on audio input. For example, the alert manager **126** may analyze the audio to determine the audio includes one or more trigger criteria and provide an alert based on the determination. To illustrate the foregoing, the client device **102** receiving audio capturing a fire alarm may trigger the alert manager **126** to alert the user to find a fire extinguisher.

(66) In at least one embodiment, the recommendation manager **128** may be used to recommend one or more locations or positions for storage and/or retrieval of the item(s) for an alert. For example, one or more recommendations may be included in the item-based alert **304**. As a further example, one or more recommendations may be generated based at least on a selection of the item-based alert **304** by a user and/or client device **102** (e.g., in a user interface **140**). In the example of a fire extinguisher, the alert manager **126** may use the recommendation manager **128** to provide a

recommendation for where the fire extinguisher is likely located (which may include providing a route to the item). Additionally or alternatively, the alert manager **126** may provide a query to the query engine **138** to search for the fire extinguisher. In one or more embodiments, once an item(s) is identified, the recommendation manager **128** and the query engine **138** may use the same components for locating the item, planning a route to the item and/or or providing a route to the item.

(67) In at least one embodiment, the client device(s) **102** may include a robot or other machine and the alert manager **126** may provide an item-based alert for one or more items to the machine (e.g., an autonomous or semi-autonomous machine, and ARM, etc.). Based on the item-based alert and/or route to an item(s) and/or the user, the machine may retrieve and/or bring the item(s) to the user. Thus, for example, a robot may preemptively bring one or more items to a user based on patterns, scheduling, and/or other triggers (e.g., receiving user requests or queries for the items).

(68) In any example, one or more users and/or client devices **102** may provide feedback regarding a location or position and/or route provided to the user (e.g., via a user interface **140**) and/or client device **102**. For example, where the query engine **138** is used to provide a location or position and/or route for the user, the user may provide feedback on whether the item was found at the location or position. In response to the feedback, the user profile manager **132** may update corresponding user data and/or the environment manager **120** may update corresponding item data. For example, if the item is indicated as found, the environment manager **120** may update the item data **136** to indicate the item was found at the location or position, which may increase the likelihood of the location or position being predicted and/or recommended in the future. If the item is indicated as not found, the environment manager **120** may update the item data **136** to indicate the item was not found at the location or position, which may decrease the likelihood of the location or position being predicted and/or recommended in the future. Additionally or alternatively, another location or position and/or route may be provided based on the indication. For example, where the locations or positions were ranked, a lower ranked location or position (e.g., having the next highest rank) may be used. This process may repeat until the item is found. Similarly, the user may provide feedback (e.g., a rating) on recommended storage locations or positions for items to impact future recommendations and/or predictions and/or to receive one or more additional recommendations for a particular session.

(69) Additionally or alternatively, the system **100** (e.g., a server **104**) may request information from one or more users and/or client devices **102** (e.g., using a user interface **140**, an alert, a notification, etc.), which may be responded to using any suitable approach described herein (e.g., using an input device **112**). For example, the system **100** may determine one or more portions of missing information in the model data **134** and/or the item data **136** (e.g., metadata) for a 3D reconstruction of an environment. Based at least on the determination, the system **100** may request information the system **100** may use to compensate for or supply the missing information. For example, where a section of a room was occluded in the sensor data **148**, the system **100** may guide the user in collecting the missing data, whether it be in the form of supplying additional sensor data used to reconstruct the missing data or otherwise (e.g., description of the missing information, answers to questions about the missing information, etc.).

(70) Now referring to FIG. 4, each block of method **400**, and other methods described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The methods may also be embodied as computer-usable instructions stored on computer storage media. The methods may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, methods are described, by way of example, with respect to FIGS. 1 and 2. However, the methods may additionally or alternatively be executed by any one system, or any combination of systems, including, but not limited to, those

described herein.

(71) FIG. 4 is a flow diagram showing the method **400** the server(s) **104** may perform for tracking and locating items based on 3D reconstruction, in accordance with some embodiments of the present disclosure.

(72) The method **400**, at block **B402**, includes reconstructing one or more objects to define a 3D representation of an environment. For example, the environment modeler **122** may reconstruct at least a portion of geometry of the objects **240** and **242** in the environment **150** using image data representing one or more videos of the environment **150** to define the 3D representation **250** of the environment **150**.

(73) The method **400**, at block **B404**, includes segmenting the one or more objects from the 3D representation. For example, the environment modeler **122** may segment the objects **240** and **242** from the 3D representation **250** of the environment **150**.

(74) The method **400**, at block **B406**, includes storing one or more items corresponding to the one or more objects in a data store. For example, the environment manager **120** may store one or more items corresponding to the objects **240** and **242** in the data store **146**. Each item of the one or more items may include including one or more attributes associated with the geometry of a corresponding object.

(75) The method **400**, at block **B408**, includes determining one or more locations or positions of an item in the 3D representation based at least on one or more attributes of the item. For example, the query engine **138** may determine, for at least an item of the one or more items, the location **162** of the item in the 3D representation **250** of the environment **150** based at least on the one or more attributes of the item.

(76) The method **400**, at block **B410**, includes planning one or more routes to at least a location (position) of the one or more locations (positions). For example, the route manager **130** may plan the route **156** to the location **162** of the one or more locations through the 3D representation **250** of the environment **150**.

(77) The method **400**, at block **B412**, includes transmitting data indicating at least a portion of the one or more routes. For example, the communication interface **114B** of the server **104** may transmit data indicating at least a portion of the routes **156** to a client device **102** (e.g., responsive to a query associated with the item).

(78) Referring now to FIG. 5, FIG. 5 is a flow diagram showing a method **500** a client device **102** may perform for tracking and locating items using a 3D reconstruction of an environment, in accordance with some embodiments of the present disclosure.

(79) The method **500**, at block **B502**, includes transmitting image data representing one or more videos of an environment, causing reconstruction of one or more objects to define a 3D representation of an environment. For example, the communication interface **114A** may transmit, to the server(s) **104**, the sensor data **148** representing one or more videos of the environment **150**. The transmitting may cause the server **104** to reconstruct at least a portion of geometry of the objects **240** and **242** in the environment **150** using the sensor data **148** to define at least a portion of the 3D representation **250** of the environment **150**.

(80) The method **500**, at block **B504**, includes transmitting first data representing a query indicating one or more items in a data store that correspond to the one or more objects. For example, the communication interface **114A** may transmit, to the server **104**, first data representing the text-based query **154** indicating one or more items in the data store **146** that correspond to the objects **240** and **242**. One or more attributes of an item of the one or more items may be associated with the geometry.

(81) The method, at block **B506**, includes receiving second data indicating at least a portion of one or more routes to one or more locations or positions of an item of the one or more items. For example, the communication interface **114A** may receive, responsive to the text-based query **154**, second data indicating at least a portion of one or more routes through the 3D representation **250** of

the environment **150** to one or more locations or positions of the item, the one or more locations or positions being based at least on the one or more attributes of the item.

(82) Referring now to FIG. **6**, FIG. **6** is a flow diagram showing a method **600** for tracking and locating items using the 3D representation **144** of the environment **150**, in accordance with some embodiments of the present disclosure.

(83) The method **600**, at block **B602**, includes generating a 3D representation of an environment. For example, the environment modeler **122** may generate the 3D representation **144** of the environment **150**.

(84) The method **600**, at block **B604**, includes storing one or more attributes of one or more items corresponding to the one or more objects identified from the 3D representation. For example, the environment manager **120** may store one or more attributes of one or more items corresponding to the one or more objects in the item data **136**.

(85) The method **600**, at block **B606**, includes determining one or more locations or positions of the one or more items based at least on the one or more attributes. For example, the query engine **138** may determine one or more locations or positions of the one or more items based at least on the one or more attributes.

(86) The method **600**, at block **B608**, includes transmitting data indicating at least a portion of one or more routes to the one or more locations or positions. For example, the communication interface **114B** may transmit data indicating at least a portion of one or more routes to the one or more locations or positions to a client device **102**.

(87) Example Computing Device

(88) FIG. **7** is a block diagram of an example computing device(s) **700** suitable for use in implementing some embodiments of the present disclosure. Computing device **700** may include an interconnect system **702** that directly or indirectly couples the following devices: memory **704**, one or more central processing units (CPUs) **706**, one or more graphics processing units (GPUs) **708**, a communication interface **710**, input/output (I/O) ports **712**, input/output components **714**, a power supply **716**, one or more presentation components **718** (e.g., display(s)), and one or more logic units **720**. In at least one embodiment, the computing device(s) **700** may comprise one or more virtual machines (VMs), and/or any of the components thereof may comprise virtual components (e.g., virtual hardware components). For non-limiting examples, one or more of the GPUs **708** may comprise one or more vGPUs, one or more of the CPUs **706** may comprise one or more vCPUs, and/or one or more of the logic units **720** may comprise one or more virtual logic units. As such, a computing device(s) **700** may include discrete components (e.g., a full GPU dedicated to the computing device **700**), virtual components (e.g., a portion of a GPU dedicated to the computing device **700**), or a combination thereof.

(89) Although the various blocks of FIG. **7** are shown as connected via the interconnect system **702** with lines, this is not intended to be limiting and is for clarity only. For example, in some embodiments, a presentation component **718**, such as a display device, may be considered an I/O component **714** (e.g., if the display is a touch screen). As another example, the CPUs **706** and/or GPUs **708** may include memory (e.g., the memory **704** may be representative of a storage device in addition to the memory of the GPUs **708**, the CPUs **706**, and/or other components). In other words, the computing device of FIG. **7** is merely illustrative. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “desktop,” “tablet,” “client device,” “mobile device,” “hand-held device,” “game console,” “electronic control unit (ECU),” “virtual reality system,” and/or other device or system types, as all are contemplated within the scope of the computing device of FIG. **7**.

(90) The interconnect system **702** may represent one or more links or busses, such as an address bus, a data bus, a control bus, or a combination thereof. The interconnect system **702** may include one or more bus or link types, such as an industry standard architecture (ISA) bus, an extended industry standard architecture (EISA) bus, a video electronics standards association (VESA) bus, a

peripheral component interconnect (PCI) bus, a peripheral component interconnect express (PCIe) bus, and/or another type of bus or link. In some embodiments, there are direct connections between components. As an example, the CPU **706** may be directly connected to the memory **704**. Further, the CPU **706** may be directly connected to the GPU **708**. Where there is direct, or point-to-point connection between components, the interconnect system **702** may include a PCIe link to carry out the connection. In these examples, a PCI bus need not be included in the computing device **700**.

(91) The memory **704** may include any of a variety of computer-readable media. The computer-readable media may be any available media that may be accessed by the computing device **700**. The computer-readable media may include both volatile and nonvolatile media, and removable and non-removable media. By way of example, and not limitation, the computer-readable media may comprise computer-storage media and communication media.

(92) The computer-storage media may include both volatile and nonvolatile media and/or removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, and/or other data types. For example, the memory **704** may store computer-readable instructions (e.g., that represent a program(s) and/or a program element(s), such as an operating system. Computer-storage media may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by computing device **700**. As used herein, computer storage media does not comprise signals per se.

(93) The computer storage media may embody computer-readable instructions, data structures, program modules, and/or other data types in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may refer to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, the computer storage media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

(94) The CPU(s) **706** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **700** to perform one or more of the methods and/or processes described herein. The CPU(s) **706** may each include one or more cores (e.g., one, two, four, eight, twenty-eight, seventy-two, etc.) that are capable of handling a multitude of software threads simultaneously. The CPU(s) **706** may include any type of processor, and may include different types of processors depending on the type of computing device **700** implemented (e.g., processors with fewer cores for mobile devices and processors with more cores for servers). For example, depending on the type of computing device **700**, the processor may be an Advanced RISC Machines (ARM) processor implemented using Reduced Instruction Set Computing (RISC) or an x86 processor implemented using Complex Instruction Set Computing (CISC). The computing device **700** may include one or more CPUs **706** in addition to one or more microprocessors or supplementary co-processors, such as math co-processors.

(95) In addition to or alternatively from the CPU(s) **706**, the GPU(s) **708** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **700** to perform one or more of the methods and/or processes described herein. One or more of the GPU(s) **708** may be an integrated GPU (e.g., with one or more of the CPU(s) **706** and/or one or more of the GPU(s) **708** may be a discrete GPU. In embodiments, one or more of the GPU(s) **708** may be a coprocessor of one or more of the CPU(s) **706**. The GPU(s) **708** may be used by the computing device **700** to render graphics (e.g., 3D graphics) or perform general purpose computations. For example, the GPU(s) **708** may be used for General-Purpose computing

on GPU(s) (GPGPU). The GPU(s) **708** may include hundreds or thousands of cores that are capable of handling hundreds or thousands of software threads simultaneously. The GPU(s) **708** may generate pixel data for output images in response to rendering commands (e.g., rendering commands from the CPU(s) **706** received via a host interface). The GPU(s) **708** may include graphics memory, such as display memory, for storing pixel data or any other suitable data, such as GPGPU data. The display memory may be included as part of the memory **704**. The GPU(s) **708** may include two or more GPUs operating in parallel (e.g., via a link). The link may directly connect the GPUs (e.g., using NVLINK) or may connect the GPUs through a switch (e.g., using NVSwitch). When combined together, each GPU **708** may generate pixel data or GPGPU data for different portions of an output or for different outputs (e.g., a first GPU for a first image and a second GPU for a second image). Each GPU may include its own memory, or may share memory with other GPUs.

(96) In addition to or alternatively from the CPU(s) **706** and/or the GPU(s) **708**, the logic unit(s) **720** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **700** to perform one or more of the methods and/or processes described herein. In embodiments, the CPU(s) **706**, the GPU(s) **708**, and/or the logic unit(s) **720** may discretely or jointly perform any combination of the methods, processes and/or portions thereof. One or more of the logic units **720** may be part of and/or integrated in one or more of the CPU(s) **706** and/or the GPU(s) **708** and/or one or more of the logic units **720** may be discrete components or otherwise external to the CPU(s) **706** and/or the GPU(s) **708**. In embodiments, one or more of the logic units **720** may be a coprocessor of one or more of the CPU(s) **706** and/or one or more of the GPU(s) **708**.

(97) Examples of the logic unit(s) **720** include one or more processing cores and/or components thereof, such as Data Processing Units (DPUs), Tensor Cores (TCs), Tensor Processing Units (TPUs), Pixel Visual Cores (PVCs), Vision Processing Units (VPUs), Graphics Processing Clusters (GPCs), Texture Processing Clusters (TPCs), Streaming Multiprocessors (SMs), Tree Traversal Units (TTUs), Artificial Intelligence Accelerators (AIAs), Deep Learning Accelerators (DLAs), Arithmetic-Logic Units (ALUs), Application-Specific Integrated Circuits (ASICs), Floating Point Units (FPUs), input/output (I/O) elements, peripheral component interconnect (PCI) or peripheral component interconnect express (PCIe) elements, and/or the like.

(98) The communication interface **710** may include one or more receivers, transmitters, and/or transceivers that enable the computing device **700** to communicate with other computing devices via an electronic communication network, included wired and/or wireless communications. The communication interface **710** may include components and functionality to enable communication over any of a number of different networks, such as wireless networks (e.g., Wi-Fi, Z-Wave, Bluetooth, Bluetooth LE, ZigBee, etc.), wired networks (e.g., communicating over Ethernet or InfiniBand), low-power wide-area networks (e.g., LoRaWAN, SigFox, etc.), and/or the Internet. In one or more embodiments, logic unit(s) **720** and/or communication interface **710** may include one or more data processing units (DPUs) to transmit data received over a network and/or through interconnect system **702** directly to (e.g., a memory of) one or more GPU(s) **708**.

(99) The I/O ports **712** may enable the computing device **700** to be logically coupled to other devices including the I/O components **714**, the presentation component(s) **718**, and/or other components, some of which may be built in to (e.g., integrated in) the computing device **700**. Illustrative I/O components **714** include a microphone, mouse, keyboard, joystick, game pad, game controller, satellite dish, scanner, printer, wireless device, etc. The I/O components **714** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, and touch recognition (as described in

more detail below) associated with a display of the computing device **700**. The computing device **700** may include depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, touchscreen technology, and combinations of these, for gesture detection and recognition. Additionally, the computing device **700** may include accelerometers or gyroscopes (e.g., as part of an inertia measurement unit (IMU)) that enable detection of motion. In some examples, the output of the accelerometers or gyroscopes may be used by the computing device **700** to render immersive augmented reality or virtual reality.

(100) The power supply **716** may include a hard-wired power supply, a battery power supply, or a combination thereof. The power supply **716** may provide power to the computing device **700** to enable the components of the computing device **700** to operate.

(101) The presentation component(s) **718** may include a display (e.g., a monitor, a touch screen, a television screen, a heads-up-display (HUD), other display types, or a combination thereof), speakers, and/or other presentation components. The presentation component(s) **718** may receive data from other components (e.g., the GPU(s) **708**, the CPU(s) **706**, DPUs, etc.), and output the data (e.g., as an image, video, sound, etc.).

(102) Example Data Center

(103) FIG. **8** illustrates an example data center **800** that may be used in at least one embodiment of the present disclosure. The data center **800** may include a data center infrastructure layer **810**, a framework layer **820**, a software layer **830**, and/or an application layer **840**.

(104) As shown in FIG. **8**, the data center infrastructure layer **810** may include a resource orchestrator **812**, grouped computing resources **814**, and node computing resources (“node C.R.s”) **816(1)-816(N)**, where “N” represents any whole, positive integer. In at least one embodiment, node C.R.s **816(1)-816(N)** may include, but are not limited to, any number of central processing units (CPUs) or other processors (including DPUs, accelerators, field programmable gate arrays (FPGAs), graphics processors or graphics processing units (GPUs), etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output (NW I/O) devices, network switches, virtual machines (VMs), power modules, and/or cooling modules, etc. In some embodiments, one or more node C.R.s from among node C.R.s **816(1)-816(N)** may correspond to a server having one or more of the above-mentioned computing resources. In addition, in some embodiments, the node C.R.s **816(1)-816(N)** may include one or more virtual components, such as vGPUs, vCPUs, and/or the like, and/or one or more of the node C.R.s **816(1)-816(N)** may correspond to a virtual machine (VM).

(105) In at least one embodiment, grouped computing resources **814** may include separate groupings of node C.R.s **816** housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node C.R.s **816** within grouped computing resources **814** may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s **816** including CPUs, GPUs, DPUs, and/or other processors may be grouped within one or more racks to provide compute resources to support one or more workloads. The one or more racks may also include any number of power modules, cooling modules, and/or network switches, in any combination.

(106) The resource orchestrator **812** may configure or otherwise control one or more node C.R.s **816(1)-816(N)** and/or grouped computing resources **814**. In at least one embodiment, resource orchestrator **812** may include a software design infrastructure (SDI) management entity for the data center **800**. The resource orchestrator **812** may include hardware, software, or some combination thereof.

(107) In at least one embodiment, as shown in FIG. **8**, framework layer **820** may include a job scheduler **833**, a configuration manager **834**, a resource manager **836**, and/or a distributed file system **838**. The framework layer **820** may include a framework to support software **832** of software layer **830** and/or one or more application(s) **842** of application layer **840**. The software

832 or application(s) **842** may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. The framework layer **820** may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter “Spark”) that may utilize distributed file system **838** for large-scale data processing (e.g., “big data”). In at least one embodiment, job scheduler **833** may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center **800**. The configuration manager **834** may be capable of configuring different layers such as software layer **830** and framework layer **820** including Spark and distributed file system **838** for supporting large-scale data processing. The resource manager **836** may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system **838** and job scheduler **833**. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource **814** at data center infrastructure layer **810**. The resource manager **836** may coordinate with resource orchestrator **812** to manage these mapped or allocated computing resources.

(108) In at least one embodiment, software **832** included in software layer **830** may include software used by at least portions of node C.R.s **816(1)-816(N)**, grouped computing resources **814**, and/or distributed file system **838** of framework layer **820**. One or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

(109) In at least one embodiment, application(s) **842** included in application layer **840** may include one or more types of applications used by at least portions of node C.R.s **816(1)-816(N)**, grouped computing resources **814**, and/or distributed file system **838** of framework layer **820**. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.), and/or other machine learning applications used in conjunction with one or more embodiments.

(110) In at least one embodiment, any of configuration manager **834**, resource manager **836**, and resource orchestrator **812** may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. Self-modifying actions may relieve a data center operator of data center **800** from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

(111) The data center **800** may include tools, services, software or other resources to train one or more machine learning models or predict or infer information using one or more machine learning models according to one or more embodiments described herein. For example, a machine learning model(s) may be trained by calculating weight parameters according to a neural network architecture using software and/or computing resources described above with respect to the data center **800**. In at least one embodiment, trained or deployed machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to the data center **800** by using weight parameters calculated through one or more training techniques, such as but not limited to those described herein.

(112) In at least one embodiment, the data center **800** may use CPUs, application-specific integrated circuits (ASICs), GPUs, FPGAs, and/or other hardware (or virtual compute resources corresponding thereto) to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

(113) Example Network Environments

(114) Network environments suitable for use in implementing embodiments of the disclosure may include one or more client devices, servers, network attached storage (NAS), other backend

devices, and/or other device types. The client devices, servers, and/or other device types (e.g., each device) may be implemented on one or more instances of the computing device(s) **700** of FIG. 7—e.g., each device may include similar components, features, and/or functionality of the computing device(s) **700**. In addition, where backend devices (e.g., servers, NAS, etc.) are implemented, the backend devices may be included as part of a data center **800**, an example of which is described in more detail herein with respect to FIG. 8.

(115) Components of a network environment may communicate with each other via a network(s), which may be wired, wireless, or both. The network may include multiple networks, or a network of networks. By way of example, the network may include one or more Wide Area Networks (WANs), one or more Local Area Networks (LANs), one or more public networks such as the Internet and/or a public switched telephone network (PSTN), and/or one or more private networks. Where the network includes a wireless telecommunications network, components such as a base station, a communications tower, or even access points (as well as other components) may provide wireless connectivity.

(116) Compatible network environments may include one or more peer-to-peer network environments—in which case a server may not be included in a network environment—and one or more client-server network environments—in which case one or more servers may be included in a network environment. In peer-to-peer network environments, functionality described herein with respect to a server(s) may be implemented on any number of client devices.

(117) In at least one embodiment, a network environment may include one or more cloud-based network environments, a distributed computing environment, a combination thereof, etc. A cloud-based network environment may include a framework layer, a job scheduler, a resource manager, and a distributed file system implemented on one or more of servers, which may include one or more core network servers and/or edge servers. A framework layer may include a framework to support software of a software layer and/or one or more application(s) of an application layer. The software or application(s) may respectively include web-based service software or applications. In embodiments, one or more of the client devices may use the web-based service software or applications (e.g., by accessing the service software and/or applications via one or more application programming interfaces (APIs)). The framework layer may be, but is not limited to, a type of free and open-source software web application framework such as that may use a distributed file system for large-scale data processing (e.g., “big data”).

(118) A cloud-based network environment may provide cloud computing and/or cloud storage that carries out any combination of computing and/or data storage functions described herein (or one or more portions thereof). Any of these various functions may be distributed over multiple locations from central or core servers (e.g., of one or more data centers that may be distributed across a state, a region, a country, the globe, etc.). If a connection to a user (e.g., a client device) is relatively close to an edge server(s), a core server(s) may designate at least a portion of the functionality to the edge server(s). A cloud-based network environment may be private (e.g., limited to a single organization), may be public (e.g., available to many organizations), and/or a combination thereof (e.g., a hybrid cloud environment).

(119) The client device(s) may include at least some of the components, features, and functionality of the example computing device(s) **700** described herein with respect to FIG. 7. By way of example and not limitation, a client device may be embodied as a Personal Computer (PC), a laptop computer, a mobile device, a smartphone, a tablet computer, a smart watch, a wearable computer, a Personal Digital Assistant (PDA), an MP3 player, a virtual reality headset, a Global Positioning System (GPS) or device, a video player, a video camera, a surveillance device or system, a vehicle, a boat, a flying vessel, a virtual machine, a drone, a robot, a handheld communications device, a hospital device, a gaming device or system, an entertainment system, a vehicle computer system, an embedded system controller, a remote control, an appliance, a consumer electronic device, a workstation, an edge device, any combination of these delineated devices, or any other suitable

device.

(120) The disclosure may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant or other handheld device.

Generally, program modules including routines, programs, objects, components, data structures, etc., refer to code that perform particular tasks or implement particular abstract data types. The disclosure may be practiced in a variety of system configurations, including hand-held devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The disclosure may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.

(121) As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.

(122) The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Claims

1. A system comprising: one or more processing units to execute operations comprising: defining, using image data of one or more videos of an environment, a simulated three-dimensional (3D) representation of at least a portion of an environment that includes a plurality of objects; determining, based at least on analyzing the simulated 3D representation of the environment, one or more areas in the simulated 3D representation having one or more geometries that are geometrically compatible with a geometry of a simulated object, the simulated object including one or more attributes that correspond to at least an object of the plurality of objects; determining, using the simulated 3D representation and based at least on the one or more areas being geometrically compatible with the geometry of the simulated object, one or more predicted locations of the object in the environment; and planning one or more routes through the environment to at least the one or more predicted locations using the simulated 3D representation of the environment.
2. The system of claim 1, wherein the one or more attributes indicate one or more of a size of the one or more objects, a shape of the one or more objects, a surface normal of the one or more objects, or an affordance of the one or more objects.
3. The system of claim 1, the operations further include comprising receiving data indicating a user selection of a storage location for the simulated object in the simulated 3D representation, wherein the determining the one or more locations is based at least on the storage location indicated by the user selection.
4. The system of claim 1, the operations further comprising: determining, for a plurality of locations in the simulated 3D representation of the environment, probability scores representing a

likelihood of the object being positioned at the locations based at least on previous storage locations of the object; and selecting the one or more predicted locations from the plurality of locations based at least on one or more first probability scores for the one or more predicted locations being greater than one or more second probability scores for one or more second predicted locations, wherein the determining of the one or more predicted locations is further based at least on the one or more first probability scores being greater than the one or more second probability scores.

5. The system of claim 1, the operations further comprising: transmitting data indicating at least a portion of the one or more routes to one or more devices, the transmitting being responsive to a query that includes data representing one or more images; analyzing the one or more videos to determine one or more features of the image; and selecting the simulated object from one or more simulated objects based at least on the one or more features and the one or more attributes, wherein the determining of the one or more areas is based at least on the selecting of the simulated object.

6. The system of claim 1, the operations further comprising: transmitting data indicating at least a portion of the one or more routes to one or more devices, the transmitting being responsive to a query that includes a natural language query; analyzing the natural language query to determine one or more features of the query; and selecting the simulated object from one or more simulated objects based at least on the one or more features and the one or more attributes, wherein the determining of the one or more areas is based at least on the selecting of the simulated object.

7. The system of claim 6, wherein the one or more devices include a machine that uses the data to navigate to the one or more predicted locations in the environment based at least on the at least a portion of the one or more routes.

8. The system of claim 6, wherein the one or more devices include a user device that uses the data to display navigation instructions corresponding to the at least a portion of the one or more routes.

9. The system of claim 1, wherein item data corresponding to at least the object of the plurality of objects is stored in a data store in association with a user profile including user data indicating one or more of previous locations of the simulated object in the simulated 3D representation of the environment, a location in the simulated 3D representation of the environment that was marked, using user input, as a current location of the simulated object, or one or more times to provide a reminder to find the simulated object.

10. The system of claim 1, wherein the planning of the one or more routes personalizes a path to the one or more predicted locations based at least on one or more user mobility characteristics associated with a user profile and a 3D geometry of the environment.

11. The system of claim 1, the operations further comprising transmitting data indicating at least a portion of the one or more routes to one or more devices, wherein the transmitting is triggered using a user recommendation algorithm that is based at least on one or more historical patterns of user requests for a user account, the user requests associated with locations for the simulated object.

12. The system of claim 1, wherein the system is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for generating or maintaining a digital twin of one or more physical objects; a system for performing simulation operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing conversational AI operations; a system for generating synthetic data; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

13. A method comprising: transmitting, to one or more servers, image data representing one or more videos of an environment, the transmitting causing the one or more servers to define at least a

portion of a simulated three-dimensional (3D) representation of the environment using the image data; transmitting, to the one or more servers, first data representing a query indicating one or more simulated objects in a data store that correspond to one or more objects in the environment, wherein one or more attributes of the one or more simulated objects correspond to the one or more objects; and receiving, responsive to the query, second data indicating at least a portion of one or more routes through the environment to one or more predicted locations of the one or more objects, the one or more predicted locations determined based at least on the one or more servers analyzing the simulated 3D representation of the environment to determine one or more areas in the simulated 3D representation that are geometrically compatible with the one or more attributes of the one or more simulated objects, the one or more areas corresponding to the one or more predicted locations.

14. The method of claim 13, wherein the image data is generated using one or more sensors of a robot, and the robot uses the second data to navigate to the one or more predicted locations based at least on the at least a portion of the one or more routes.

15. The method of claim 13, wherein the image data is generated using one or more sensors of a user device, and the user device uses the second data to display navigation instructions to the one or more predicted locations in the environment based at least on the at least a portion of the one or more routes.

16. The method of claim 13, wherein the query includes one or more of data representing one or more images that at least partially depict a representation of the one or more simulated objects or a natural language query associated with the one or more simulated objects.

17. A processor comprising: one or more circuits to generate a simulated three-dimensional (3D) representation of an environment using image data depicting at least a portion of the environment, store one or more attributes of one or more simulated objects corresponding to one or more objects identified from the simulated 3D representation of the environment in a data store, determine one or more predicted locations of the one or more objects in the environment based at least on analyzing the simulated 3D representation to determine one or more areas that are geometrically compatible with the one or more attributes of the one or more simulated objects, the one or more areas corresponding to the one or more predicted locations, and transmit data indicating at least a portion of one or more routes through the environment to the one or more predicted locations.

18. The processor of claim 17, wherein the one or more attributes indicate one or more of a size of the one or more objects, a shape of the one or more objects, a surface normal of the one or more objects, or an affordance of the one or more objects.

19. The processor of claim 17, wherein the one or more circuits are further to receive data indicating a user selection of one or more storage locations for the one or more simulated objects in the simulated 3D representation, wherein the determining the one or more predicted locations is based at least on the one or more storage locations indicated by the user selection.

20. The processor of claim 17, wherein the determining the one or more predicted locations is based at least on: determining a probability corresponding to a likelihood the one or more simulated objects correspond to a location of the one or more predicted locations based at least on one or more previous storage locations of the one or more simulated objects; and selecting the one or more predicted locations based at least on the probability, wherein the transmission of the data is based at least on the selecting of the one or more predicted locations.

21. The processor of claim 17, wherein the processor is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for generating or maintaining a digital twin of one or more physical objects; a system for performing simulation operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing conversational AI operations; a system for

generating synthetic data; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

22. The system of claim 1, wherein the analyzing of the simulated 3D representation of the environment comprises: comparing, to the one or more geometries associated with the one or more areas, a 3D model representative of the geometry of the simulated object; and determining, based at least on the comparing, that the one or more areas are geometrically compatible with the simulated object such that the simulated object fits within the one or more areas.

23. The processor of claim 17, wherein the one or more circuits are further to: determine that the one or more simulated objects fit within the one or more areas, wherein the determination of the one or more areas that are geometrically compatible with the one or more attributes of the one or more simulated objects is based at least on the determination that the one or more simulated objects fit within the one or more areas.
